

UTILITY

Section A

Duration of each question is **40 mins**.

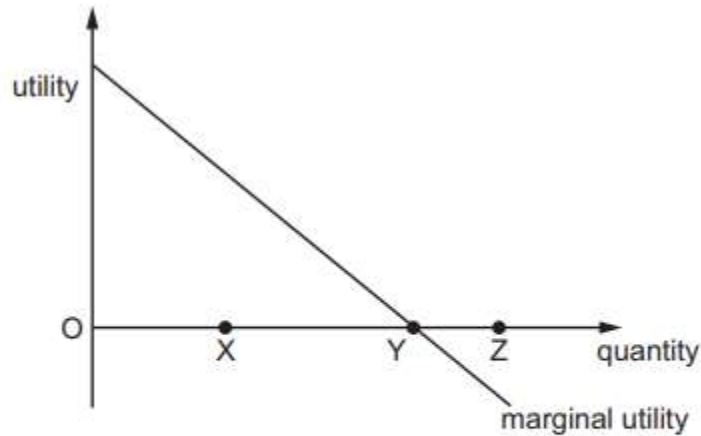
- 2** Evaluate whether marginal utility theory can fully explain the link between the changing price of a good and quantity demanded of that good. [20]

Section B

Answer **all** questions

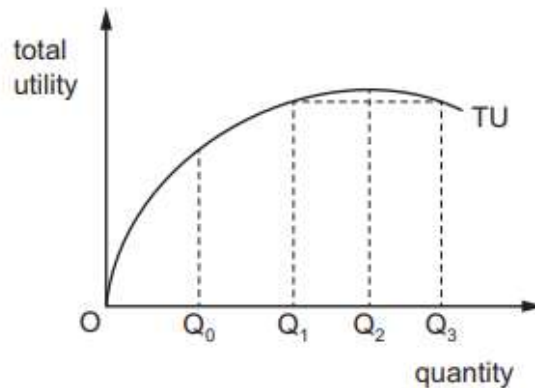
Duration 1hr 15mins

1. What does the marginal utility diagram show?



- A** Total utility becomes negative if the quantity consumed increases from Y to Z.
 - B** Total utility decreases if the quantity consumed increases from X to Y.
 - C** Total utility decreases if the quantity consumed increases from Y to Z.
 - D** Total utility is maximised at X.
2. What is the central principle of marginal utility theory?
- A** As more is consumed of a normal good, the additional satisfaction gained increases.
 - B** Consumers are in equilibrium when the marginal utility gained from good X equals the price of good Y.
 - C** Consumers are in equilibrium when the ratio of the marginal utility of goods X and Y are equal to the ratio of their prices.
 - D** Consumers are in equilibrium when the total utility gained from good X is the same as gained from good Y.

3. The diagram shows the total utility of a good for an individual consumer. Marginal utility gives a guide to the price a consumer is willing to pay.



What can be concluded about the price a consumer is willing to pay per unit?

- A The price a consumer is willing to pay will be higher at Q_0 than Q_1 .
 - B The price a consumer is willing to pay will be higher at Q_3 than Q_0 .
 - C The price a consumer is willing to pay will be highest at Q_2 .
 - D The price a consumer is willing to pay will be the same at Q_1 and Q_3 .
4. A consumer maximises his utility by spending all his income on goods X and Y. New research reveals that good X is a merit good providing the consumer with more utility than was previously realised. The consumer's income and the prices of goods X and Y are unchanged.

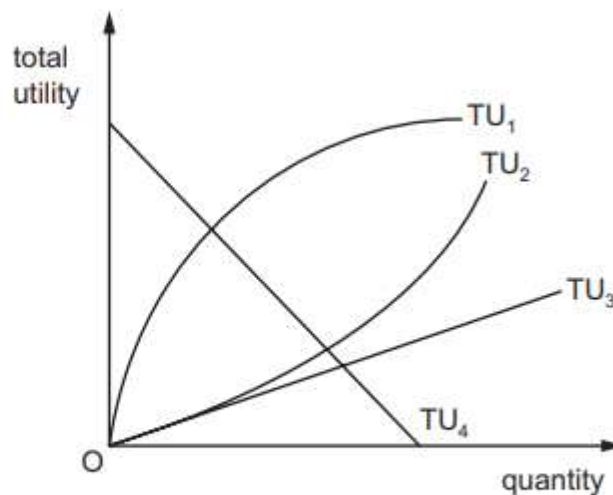
If the consumer behaves rationally, how will he respond to the findings of the research?

	quantity of good X purchased	quantity of good Y purchased
A	decreases	increases
B	increases	decreases
C	increases	unchanged
D	unchanged	increases

5. Which act of purchase or consumption is most likely to illustrate the law of diminishing marginal utility?

- A** the final course of an 'eat all you can' banquet to end an evening out
- B** the last coin to complete an enthusiast's coin collection
- C** the second of the two rolls of wallpaper needed to decorate a room
- D** the tenth driving lesson needed to enable a learner to pass the driving test

6. The diagram shows a consumer's total utility.



Which curve would mean that marginal utility is diminishing?

- A** TU_1
- B** TU_2
- C** TU_3
- D** TU_4

7. What is it called when a consumer's marginal utility is greater than the price paid for the good?

- A** a Giffen good
- B** an inferior good
- C** consumer surplus
- D** producer surplus

8. If consumers aim to maximise their utility, how will they arrange their spending?

- A** to obtain the same total utility from each commodity
- B** to obtain the same total utility per \$ spent on each commodity
- C** to obtain the same utility from the last unit of each commodity
- D** to obtain the same utility from the last \$ spent on each commodity

9. A consumer currently consumes 10 units of good X and 10 units of good Y at point Z on an indifference curve.

The consumer's marginal utilities from consuming both good X and good Y are positive.

Which combination of good X and good Y could be on the same indifference curve as point Z?

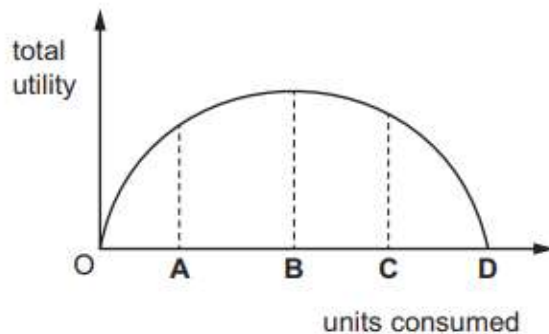
	units of good X	units of good Y
A	9	11
B	10	11
C	11	10
D	12	12

10. What is held constant when deriving an individual consumer's downward-sloping demand curve for a product?

- A** marginal utility
- B** the consumer's income
- C** the price of the product
- D** total utility

11. The diagram shows an individual's total utility curve as different quantities of a good are consumed.

At which level of consumption does the individual's marginal utility become zero?



12. Why does a normal demand curve for a product slope downwards from left to right?
- A Buyers' additional satisfaction declines as consumption rises.
 - B Consumers are faced with choices between competing products.
 - C Sellers are willing to accept lower prices on larger orders.
 - D The average cost of production falls as the scale of production increases.
13. To maximise the satisfaction he derives from a given level of expenditure on two goods, X and Y, a consumer should allocate his expenditure between the two goods so that
- A marginal utility of X = price of X and marginal utility of Y = price of Y.
 - B marginal utility of X plus marginal utility of Y is maximised.
 - C marginal utility of X = marginal utility of Y.
 - D $\frac{\text{marginal utility of X}}{\text{marginal utility of Y}} = \frac{\text{price of X}}{\text{price of Y}}$.

14. What will happen if, in response to a fall in its price, a consumer buys more of a good?

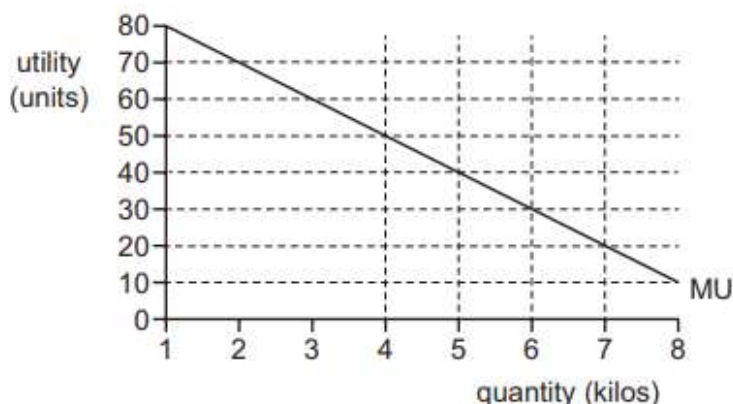
- A a fall in the total utility derived from consuming the good
- B a movement along the consumer's demand curve
- C a rise in the marginal utility derived from consuming the good
- D a shift in the consumer's demand curve

15. A rational consumer maximises satisfaction by buying three units of good X and four units of good Y which are the same dollar price.

What does this indicate?

- A Marginal utility per dollar spent is equal for both goods.
- B The total utility of 3X is equal to the total utility of 4Y.
- C The consumer should buy one more unit of good X to maximise utility.
- D Good Y has greater marginal utility than good X.

16. The diagram shows the marginal utility (MU) that an individual derives from a good at different levels of consumption.



The utility he derives from the last \$ he spends on every good is 4 units.

Assuming the marginal utility of money is constant, which quantity will he purchase if the price of the good is \$10?

- A 4 kilos
- B 5 kilos
- C 6 kilos
- D 7 kilos

17. When does a consumer achieve maximum total utility (TU), according to marginal utility (MU) theory for two goods, X and Y, with different prices (P)?

- A $MU_X = MU_Y$
- B $\frac{MU_X}{P_X} = \frac{MU_Y}{P_Y}$
- C $\frac{MU_X}{P_Y} = \frac{MU_Y}{P_X}$
- D $TU_X = TU_Y$

18. In the UK in 2015 there were two methods of charging for water supply.

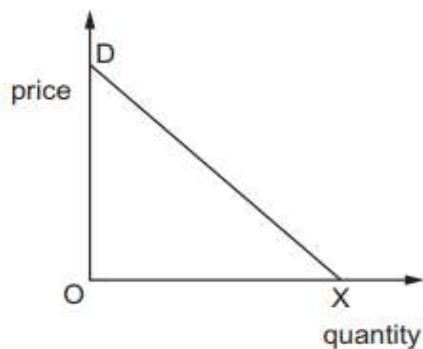
- 1 a fixed charge giving the consumer the right to consume water at zero price per litre
- 2 a price per litre of water used

It has been observed that the amount of water consumed is markedly lower when method 2 is used.

What is implied by this observation?

- A Water in the UK is not a scarce good.
- B Water is a free good of nature and no charge should be levied.
- C Water is a necessity, the use of which should not be restricted by unit prices.
- D Water is subject to the law of diminishing marginal utility.

19. The diagram shows the demand curve for a product.



What is maximised at X?

- A** marginal revenue
 - B** total revenue
 - C** marginal utility
 - D** total utility
20. At major sporting events, unofficial markets occur where tickets originally sold by the event organiser are resold at prices higher than the original ticket price.

What might be the conclusion of an economist considering these markets?

- A** Allocation of tickets by ballot would be preferable.
- B** The market for the resold tickets is more equitable.
- C** The new allocation of tickets better reflects consumers' utility.
- D** This is a case of monopoly exploitation.

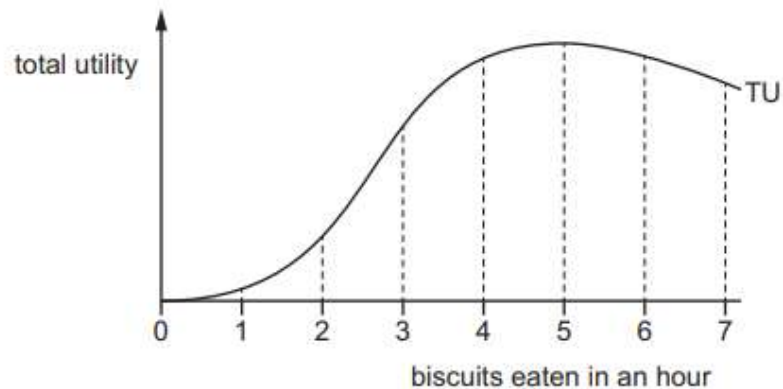
21. The table shows the total utility gained by a consumer as they consume more apples.

apples consumed	total utility
0	0
1	10
2	18
3	24
4	28
5	30
6	30

At which level of consumption would the marginal utility be zero?

- A** 1 **B** 4 **C** 5 **D** 6

22. The graph shows the total utility (TU) obtained by a consumer from eating biscuits over a period of one hour.



After eating how many biscuits does the consumer's marginal utility begin to diminish?

- A** 3 **B** 4 **C** 5 **D** 6

23. Rational consumer Romesh spent 90 rupees buying herb X and spice Y. Both cost 10 rupees per gram.

grams purchased	total utility herb X	total utility spice Y
2	15	12
3	30	25
4	45	35
5	55	42

How much of each should he buy?

	herb X (grams)	spice Y (grams)
A	2	3
B	3	5
C	4	5
D	5	4

24. Donald maximises his utility when buying paperback books and audio books. The price of a paperback book = \$3 and the price of an audio book = \$6.

If the marginal utility of the last paperback book was 12 units of utility (utils), what was the marginal utility of the last audio book purchased?

- A** 2 utils **B** 12 utils **C** 16 utils **D** 24 utils

25. A consumer has \$3 to spend each day on chocolate bars or toffee bars, which all cost \$1 each. The table below shows the units of additional satisfaction she gets from consuming each bar.

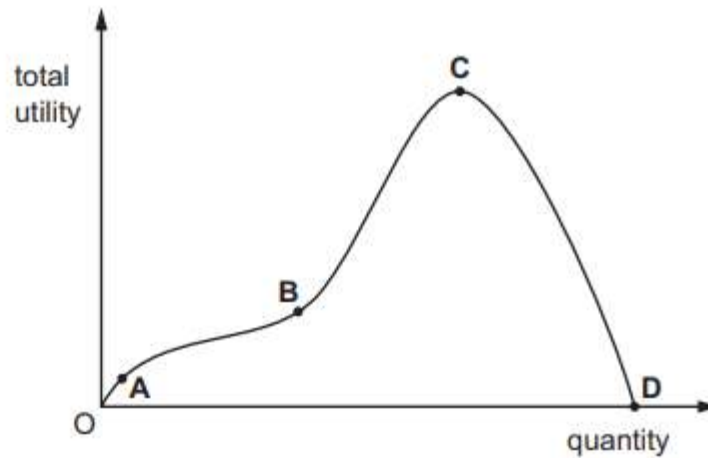
1 st chocolate bar 16	2 nd chocolate bar 13	3 rd chocolate bar 10
1 st toffee bar 14	2 nd toffee bar 12	3 rd toffee bar 9

What should she buy to maximise her total utility?

- A 1 chocolate bar and 2 toffee bars
 - B 2 chocolate bars and 1 toffee bar
 - C 3 chocolate bars
 - D 3 toffee bars
26. A rational consumer chooses what quantities of two products Y and Z to purchase with a given income.
- MU_Y and MU_Z are the additions to total utility that would result if the consumer were to purchase an additional unit of each product.
- P_Y and P_Z are the current prices of the two products.
- Which outcome would represent consumer equilibrium?
- A when $MU_Y = MU_Z$
 - B when $MU_Y \times P_Y = MU_Z \times P_Z$
 - C when $MU_Y \times P_Z = MU_Z \times P_Y$
 - D it is not possible to tell from the information available

27. The diagram shows a total utility curve for a consumer.

At which point does marginal utility equal zero?



28. The table contains data on the total utility gained by a consumer from eating biscuits and chocolates in one day.

biscuits / chocolates eaten	total utility (units)	
	biscuits	chocolates
1	20	10
2	38	20
3	52	32
4	62	38
5	70	42
6	74	42
7	74	40

What can be concluded from the table?

- A** The consumer gets diminishing marginal utility after eating the first chocolate.
- B** The consumer gets negative marginal utility from the seventh chocolate eaten.
- C** The consumer will never eat chocolates if biscuits are available.
- D** The second biscuit eaten gives the same marginal utility as the fourth chocolate eaten.

29. An individual consumes goods X and Y. The marginal utility of good X is positive and falling. The consumer is at the point where the ratio of the marginal utility of the two goods is equal to the ratio of their prices.

What can be concluded from this information?

- A the consumer will continue to consume additional units of X until marginal utility is zero
 - B the consumer will purchase more of X if the price of Y rises
 - C total utility will be negative on the additional unit of X consumed
 - D total utility will be rising at an increasing rate on the additional unit of X consumed
30. To maximise the satisfaction she derives from a given level of expenditure on two goods, X and Y, how should a consumer allocate her expenditure between the two goods?
- A $\text{marginal utility X} = \text{marginal utility Y}$
 - B $\text{marginal utility X} \times \text{price of X} = \text{marginal utility Y}$
 - C $\frac{\text{marginal utility X}}{\text{price of Y}} = \frac{\text{marginal utility Y}}{\text{price of X}}$
 - D $\frac{\text{marginal utility X}}{\text{marginal utility Y}} = \frac{\text{price of X}}{\text{price of Y}}$

UTILITY

(MARKING SCHEME)

Question	Answer	Marks
2	Evaluate whether marginal utility theory can fully explain the link between the changing price of a good and quantity demanded of that good. Use Table A: AO1 Knowledge and understanding and AO2 Analysis and Table B: AO3 Evaluation to mark candidate responses to this question. AO1 and AO2 out of 14 marks. AO3 out of 6 marks. Indicative content Responses may include: AO1 Knowledge and understanding and AO2 Analysis • Definitions of marginal and total utility. • Explanation of total utility and diminishing marginal utility. • Analysis of the effect of a change in price through the equi-marginal principle and illustration of the effect with a diagram. • Diagram of the falling demand curve v price, which can apply to both normal and inferior (non-Giffen) goods but the theory does not distinguish between them. AO3 Evaluation • Discussion of the validity of the rational consumer. Imperfect information, bounded rationality, impact of advertising. • Unable to distinguish between the income and substitution effects and therefore does not account for Giffen goods. • The difficulty of the ceteris paribus assumption that other things do not change when in practice they do. Each change would require a re-estimation of the demand schedule. • The assumption that money has constant utility. Is the last dollar of someone earning \$10 000 the same value as someone earning \$100 000. • Difficulty of MU theory accounting for the purchase of a one-off good e.g. a house. Accept all valid responses.	20

UTILITY
A2 LEVEL
ECONOMICS P3

1.	C	11.	B	21.	D
2.	C	12.	A	22.	A
3.	A	13.	D	23.	D
4.	B	14.	B	24.	D
5.	A	15.	A	25.	B
6.	A	16.	B	26.	C
7.	C	17.	B	27.	C
8.	D	18.	D	28.	B
9.	A	19.	D	29.	B
10.	B	20.	C	30.	D